

## ANALYTICAL REPORT

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Laboratory Job ID: 190-21362-1

Client Project/Site: WWTP Semiannual PFAS

For:

City of Cadillac Utilities  
1121 Plett Road  
Cadillac, Michigan 49601

Attn: Cindy Tomaszewski

*Sue Schafer*

Authorized for release by:  
11/14/2019 7:49:21 AM

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*Results relate only to the items tested and the sample(s) as received by the laboratory.*

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## Sample Summary

Client: City of Cadillac Utilities  
Project/Site: WWTP Semiannual PFAS

Job ID: 190-21362-1

| Lab Sample ID | Client Sample ID   | Matrix | Collected      | Received       | Asset ID |
|---------------|--------------------|--------|----------------|----------------|----------|
| 190-21362-1   | Field Blank        | Water  | 10/22/19 11:50 | 10/23/19 13:02 |          |
| 190-21362-2   | Effluent           | Water  | 10/22/19 11:55 | 10/23/19 13:02 |          |
| 190-21362-3   | Effluent Duplicate | Water  | 10/22/19 11:55 | 10/23/19 13:02 |          |

# Case Narrative

Client: City of Cadillac Utilities  
Project/Site: WWTP Semiannual PFAS

Job ID: 190-21362-1

**Job ID: 190-21362-1**

**Laboratory: Eurofins TestAmerica, Michigan**

## Narrative

### Job Narrative 190-21362-1

#### Comments

No additional comments.

#### Receipt

The samples were received on 10/23/2019 1:02 PM; the samples arrived in good condition, properly preserved and, where required, on ice. The temperature of the cooler at receipt was 2.2° C.

#### LCMS

Method 537 (modified): M2-4:2 FTS Isotope Dilution Analyte (IDA) recovery is above the method recommended limit for the following sample: Effluent Duplicate (190-21362-3). Quantitation by isotope dilution generally precludes any adverse effect on data quality due to elevated IDA recoveries.

Method 537 (modified): 13C2 PFTeDA Isotope Dilution Analyte (IDA) recovery associated with the following sample is below the method recommended limit: Effluent Duplicate (190-21362-3). Generally, data quality is not considered affected if the IDA signal-to-noise ratio is greater than 10:1, which is achieved for all IDA in the sample(s). All detection limits are below the lower calibration.

Method 537 (modified): M2-4:2 FTS and M2-8:2 FTS Isotope Dilution Analyte (IDA) recoveries are above the method recommended limit for the following sample: Effluent (190-21362-2). Quantitation by isotope dilution generally precludes any adverse effect on data quality due to elevated IDA recoveries.

No additional analytical or quality issues were noted, other than those described above or in the Definitions/Glossary page.

# Client Sample Results

Client: City of Cadillac Utilities  
Project/Site: WWTP Semiannual PFAS

Job ID: 190-21362-1

Client Sample ID: Field Blank

Lab Sample ID: 190-21362-1

Date Collected: 10/22/19 11:50

Matrix: Water

Date Received: 10/23/19 13:02

## Method: 537 (modified) - Fluorinated Alkyl Substances

| Analyte                                                  | Result | Qualifier | RL  | Unit | D | Prepared       | Analyzed       | Dil Fac |
|----------------------------------------------------------|--------|-----------|-----|------|---|----------------|----------------|---------|
| 4:2 FTS                                                  | <19    |           | 19  | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| 6:2 FTS                                                  | <19    |           | 19  | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| 8:2 FTS                                                  | <19    |           | 19  | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)  | <19    |           | 19  | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA) | <19    |           | 19  | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| Perfluorobutanesulfonic acid (PFBS)                      | <1.9   |           | 1.9 | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| Perfluorobutanoic acid (PFBA)                            | <1.9   |           | 1.9 | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| Perfluorodecanesulfonic acid (PFDS)                      | <1.9   |           | 1.9 | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| Perfluorodecanoic acid (PFDA)                            | <1.9   |           | 1.9 | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| Perfluorododecanoic acid (PFDoA)                         | <1.9   |           | 1.9 | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| Perfluoroheptanesulfonic Acid (PFHpS)                    | <1.9   |           | 1.9 | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| Perfluoroheptanoic acid (PFHpA)                          | <1.9   |           | 1.9 | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| Perfluorohexanesulfonic acid (PFHxS)                     | <1.9   |           | 1.9 | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| Perfluorohexanoic acid (PFHxA)                           | <1.9   |           | 1.9 | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| Perfluorononanesulfonic acid (PFNS)                      | <1.9   |           | 1.9 | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| Perfluorononanoic acid (PFNA)                            | <1.9   |           | 1.9 | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| Perfluorooctanesulfonamide (FOSA)                        | <9.4   |           | 9.4 | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| Perfluorooctane sulfonate (PFOS)                         | <1.9   |           | 1.9 | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| Perfluorooctanoic acid (PFOA)                            | <1.9   |           | 1.9 | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| Perfluoropentanesulfonic acid (PFPeS)                    | <1.9   |           | 1.9 | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| Perfluoropentanoic acid (PFPeA)                          | <1.9   |           | 1.9 | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| Perfluorotetradecanoic acid (PFTeA)                      | <1.9   |           | 1.9 | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| Perfluorotridecanoic acid (PFTrIA)                       | <1.9   |           | 1.9 | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| Perfluoroundecanoic acid (PFUnA)                         | <1.9   |           | 1.9 | ng/L |   | 10/31/19 12:32 | 11/07/19 14:22 | 1       |

| Isotope Dilution | %Recovery | Qualifier | Limits   | Prepared       | Analyzed       | Dil Fac |
|------------------|-----------|-----------|----------|----------------|----------------|---------|
| 13C8 FOSA        | 92        |           | 25 - 150 | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| 13C4 PFBA        | 102       |           | 25 - 150 | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| 13C2 PFDA        | 102       |           | 50 - 150 | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| 13C2 PFDoA       | 92        |           | 50 - 150 | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| 13C4 PFHpA       | 105       |           | 50 - 150 | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| 13C2 PFHxA       | 105       |           | 50 - 150 | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| 13C5 PFNA        | 104       |           | 50 - 150 | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| 13C4 PFOA        | 101       |           | 50 - 150 | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| 13C4 PFOS        | 111       |           | 50 - 150 | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| 13C5-PFPeA DNU   | 105       |           | 25 - 150 | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| 13C2 PFTeDA      | 84        |           | 50 - 150 | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| 13C2 PFUnA       | 93        |           | 50 - 150 | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| d5-NEtFOSAA      | 86        |           | 50 - 150 | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| d3-NMeFOSAA      | 86        |           | 50 - 150 | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| M2-4:2 FTS       | 113       |           | 25 - 150 | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| M2-6:2 FTS       | 109       |           | 25 - 150 | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| M2-8:2 FTS       | 118       |           | 25 - 150 | 10/31/19 12:32 | 11/07/19 14:22 | 1       |
| 18O2 PFHxS       | 109       |           | 50 - 150 | 10/31/19 12:32 | 11/07/19 14:22 | 1       |

Eurofins TestAmerica, Michigan

# Client Sample Results

Client: City of Cadillac Utilities  
Project/Site: WWTP Semiannual PFAS

Job ID: 190-21362-1

Client Sample ID: Effluent

Lab Sample ID: 190-21362-2

Date Collected: 10/22/19 11:55

Matrix: Water

Date Received: 10/23/19 13:02

## Method: 537 (modified) - Fluorinated Alkyl Substances

| Analyte                                                  | Result    | Qualifier | RL       | Unit | D | Prepared       | Analyzed       | Dil Fac |
|----------------------------------------------------------|-----------|-----------|----------|------|---|----------------|----------------|---------|
| 4:2 FTS                                                  | <16       |           | 16       | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| 6:2 FTS                                                  | <16       |           | 16       | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| 8:2 FTS                                                  | <16       |           | 16       | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)  | <16       |           | 16       | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA) | <16       |           | 16       | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| Perfluorobutanesulfonic acid (PFBS)                      | 18        |           | 1.6      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| Perfluorobutanoic acid (PFBA)                            | 3.5       |           | 1.6      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| Perfluorodecanesulfonic acid (PFDS)                      | <1.6      |           | 1.6      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| Perfluorodecanoic acid (PFDA)                            | <1.6      |           | 1.6      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| Perfluorododecanoic acid (PFDoA)                         | <1.6      |           | 1.6      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| Perfluoroheptanesulfonic Acid (PFHpS)                    | <1.6      |           | 1.6      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| Perfluoroheptanoic acid (PFHpA)                          | <1.6      |           | 1.6      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| Perfluorohexanesulfonic acid (PFHxS)                     | 2.1       |           | 1.6      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| Perfluorohexanoic acid (PFHxA)                           | 17        |           | 1.6      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| Perfluorononanesulfonic acid (PFNS)                      | <1.6      |           | 1.6      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| Perfluorononanoic acid (PFNA)                            | <1.6      |           | 1.6      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| Perfluorooctanesulfonamide (FOSA)                        | <8.2      |           | 8.2      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| Perfluorooctane sulfonate (PFOS)                         | 2.0       |           | 1.6      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| Perfluorooctanoic acid (PFOA)                            | 3.4       |           | 1.6      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| Perfluoropentanesulfonic acid (PFPeS)                    | <1.6      |           | 1.6      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| Perfluoropentanoic acid (PFPeA)                          | 6.3       |           | 1.6      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| Perfluorotetradecanoic acid (PFTeA)                      | <1.6      |           | 1.6      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| Perfluorotridecanoic acid (PFTriA)                       | <1.6      |           | 1.6      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| Perfluoroundecanoic acid (PFUnA)                         | <1.6      |           | 1.6      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| Isotope Dilution                                         | %Recovery | Qualifier | Limits   |      |   | Prepared       | Analyzed       | Dil Fac |
| 13C8 FOSA                                                | 85        |           | 25 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| 13C4 PFBA                                                | 104       |           | 25 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| 13C2 PFDA                                                | 100       |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| 13C2 PFDoA                                               | 82        |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| 13C4 PFHpA                                               | 107       |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| 13C2 PFHxA                                               | 109       |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| 13C5 PFNA                                                | 103       |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| 13C4 PFOA                                                | 101       |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| 13C4 PFOS                                                | 90        |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| 13C5-PFPeA DNU                                           | 101       |           | 25 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| 13C2 PFTeDA                                              | 52        |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| 13C2 PFUnA                                               | 93        |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| d5-NEtFOSAA                                              | 99        |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| d3-NMeFOSAA                                              | 83        |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| M2-4:2 FTS                                               | 192 *     |           | 25 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| M2-6:2 FTS                                               | 143       |           | 25 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| M2-8:2 FTS                                               | 162 *     |           | 25 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |
| 18O2 PFHxS                                               | 89        |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:36 | 1       |

Eurofins TestAmerica, Michigan

# Client Sample Results

Client: City of Cadillac Utilities  
Project/Site: WWTP Semiannual PFAS

Job ID: 190-21362-1

Client Sample ID: Effluent Duplicate

Lab Sample ID: 190-21362-3

Date Collected: 10/22/19 11:55

Matrix: Water

Date Received: 10/23/19 13:02

## Method: 537 (modified) - Fluorinated Alkyl Substances

| Analyte                                                  | Result    | Qualifier | RL       | Unit | D | Prepared       | Analyzed       | Dil Fac |
|----------------------------------------------------------|-----------|-----------|----------|------|---|----------------|----------------|---------|
| 4:2 FTS                                                  | <17       |           | 17       | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| 6:2 FTS                                                  | <17       |           | 17       | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| 8:2 FTS                                                  | <17       |           | 17       | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)  | <17       |           | 17       | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA) | <17       |           | 17       | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| Perfluorobutanesulfonic acid (PFBS)                      | 16        |           | 1.7      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| Perfluorobutanoic acid (PFBA)                            | 3.5       |           | 1.7      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| Perfluorodecanesulfonic acid (PFDS)                      | <1.7      |           | 1.7      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| Perfluorodecanoic acid (PFDA)                            | <1.7      |           | 1.7      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| Perfluorododecanoic acid (PFDoA)                         | <1.7      |           | 1.7      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| Perfluoroheptanesulfonic Acid (PFHpS)                    | <1.7      |           | 1.7      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| Perfluoroheptanoic acid (PFHpA)                          | <1.7      |           | 1.7      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| Perfluorohexanesulfonic acid (PFHxS)                     | 2.0       |           | 1.7      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| Perfluorohexanoic acid (PFHxA)                           | 17        |           | 1.7      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| Perfluorononanesulfonic acid (PFNS)                      | <1.7      |           | 1.7      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| Perfluorononanoic acid (PFNA)                            | <1.7      |           | 1.7      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| Perfluorooctanesulfonamide (FOSA)                        | <8.3      |           | 8.3      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| Perfluorooctane sulfonate (PFOS)                         | 2.4       |           | 1.7      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| Perfluorooctanoic acid (PFOA)                            | 3.2       |           | 1.7      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| Perfluoropentanesulfonic acid (PFPeS)                    | <1.7      |           | 1.7      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| Perfluoropentanoic acid (PFPeA)                          | 6.0       |           | 1.7      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| Perfluorotetradecanoic acid (PFTeA)                      | <1.7      |           | 1.7      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| Perfluorotridecanoic acid (PFTriA)                       | <1.7      |           | 1.7      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| Perfluoroundecanoic acid (PFUnA)                         | <1.7      |           | 1.7      | ng/L |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| Isotope Dilution                                         | %Recovery | Qualifier | Limits   |      |   | Prepared       | Analyzed       | Dil Fac |
| 13C8 FOSA                                                | 78        |           | 25 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| 13C4 PFBA                                                | 100       |           | 25 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| 13C2 PFDA                                                | 88        |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| 13C2 PFDoA                                               | 77        |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| 13C4 PFHpA                                               | 102       |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| 13C2 PFHxA                                               | 104       |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| 13C5 PFNA                                                | 90        |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| 13C4 PFOA                                                | 97        |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| 13C4 PFOS                                                | 94        |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| 13C5-PFPeA DNU                                           | 98        |           | 25 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| 13C2 PFTeDA                                              | 48 *      |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| 13C2 PFUnA                                               | 80        |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| d5-NEtFOSAA                                              | 90        |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| d3-NMeFOSAA                                              | 89        |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| M2-4:2 FTS                                               | 170 *     |           | 25 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| M2-6:2 FTS                                               | 148       |           | 25 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| M2-8:2 FTS                                               | 149       |           | 25 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |
| 18O2 PFHxS                                               | 94        |           | 50 - 150 |      |   | 10/31/19 12:32 | 11/02/19 00:44 | 1       |

Eurofins TestAmerica, Michigan

# QC Sample Results

Client: City of Cadillac Utilities  
Project/Site: WWTP Semiannual PFAS

Job ID: 190-21362-1

## Method: 537 (modified) - Fluorinated Alkyl Substances

Lab Sample ID: MB 200-149141/1-A

Matrix: Water

Analysis Batch: 149190

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 149141

| Analyte                                                  | MB Result | MB Qualifier | RL  | Unit | D | Prepared       | Analyzed       | Dil Fac |
|----------------------------------------------------------|-----------|--------------|-----|------|---|----------------|----------------|---------|
| 4:2 FTS                                                  | <20       |              | 20  | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| 6:2 FTS                                                  | <20       |              | 20  | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| 8:2 FTS                                                  | <20       |              | 20  | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)  | <20       |              | 20  | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA) | <20       |              | 20  | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| Perfluorobutanesulfonic acid (PFBS)                      | <2.0      |              | 2.0 | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| Perfluorobutanoic acid (PFBA)                            | <2.0      |              | 2.0 | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| Perfluorodecanesulfonic acid (PFDS)                      | <2.0      |              | 2.0 | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| Perfluorodecanoic acid (PFDA)                            | <2.0      |              | 2.0 | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| Perfluorododecanoic acid (PFDoA)                         | <2.0      |              | 2.0 | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| Perfluoroheptanesulfonic Acid (PFHpS)                    | <2.0      |              | 2.0 | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| Perfluoroheptanoic acid (PFHpA)                          | <2.0      |              | 2.0 | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| Perfluorohexanesulfonic acid (PFHxS)                     | <2.0      |              | 2.0 | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| Perfluorohexanoic acid (PFHxA)                           | <2.0      |              | 2.0 | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| Perfluorononanesulfonic acid (PFNS)                      | <2.0      |              | 2.0 | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| Perfluorononanoic acid (PFNA)                            | <2.0      |              | 2.0 | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| Perfluorooctanesulfonamide (FOSA)                        | <10       |              | 10  | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| Perfluorooctane sulfonate (PFOS)                         | <2.0      |              | 2.0 | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| Perfluorooctanoic acid (PFOA)                            | <2.0      |              | 2.0 | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| Perfluoropentanesulfonic acid (PFPeS)                    | <2.0      |              | 2.0 | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| Perfluoropentanoic acid (PFPeA)                          | <2.0      |              | 2.0 | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| Perfluorotetradecanoic acid (PFTeA)                      | <2.0      |              | 2.0 | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| Perfluorotridecanoic acid (PFTriA)                       | <2.0      |              | 2.0 | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| Perfluoroundecanoic acid (PFUnA)                         | <2.0      |              | 2.0 | ng/L |   | 10/31/19 12:32 | 11/01/19 21:27 | 1       |

| Isotope Dilution | MB %Recovery | MB Qualifier | Limits   | Prepared       | Analyzed       | Dil Fac |
|------------------|--------------|--------------|----------|----------------|----------------|---------|
| 13C8 FOSA        | 71           |              | 25 - 150 | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| 13C4 PFBA        | 100          |              | 25 - 150 | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| 13C2 PFDA        | 102          |              | 50 - 150 | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| 13C2 PFDoA       | 77           |              | 50 - 150 | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| 13C4 PFHpA       | 102          |              | 50 - 150 | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| 13C2 PFHxA       | 104          |              | 50 - 150 | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| 13C5 PFNA        | 106          |              | 50 - 150 | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| 13C4 PFOA        | 107          |              | 50 - 150 | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| 13C4 PFOS        | 109          |              | 50 - 150 | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| 13C5-PFPeA DNU   | 106          |              | 25 - 150 | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| 13C2 PFTeDA      | 67           |              | 50 - 150 | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| 13C2 PFUnA       | 87           |              | 50 - 150 | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| d5-NEtFOSAA      | 84           |              | 50 - 150 | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| d3-NMeFOSAA      | 86           |              | 50 - 150 | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| M2-4:2 FTS       | 104          |              | 25 - 150 | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| M2-6:2 FTS       | 114          |              | 25 - 150 | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| M2-8:2 FTS       | 119          |              | 25 - 150 | 10/31/19 12:32 | 11/01/19 21:27 | 1       |
| 18O2 PFHxS       | 104          |              | 50 - 150 | 10/31/19 12:32 | 11/01/19 21:27 | 1       |

Eurofins TestAmerica, Michigan

# QC Sample Results

Client: City of Cadillac Utilities  
Project/Site: WWTP Semiannual PFAS

Job ID: 190-21362-1

## Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCS 200-149141/2-A

Matrix: Water

Analysis Batch: 149190

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 149141

| Analyte                                                  | Spike Added | LCS Result | LCS Qualifier | Unit | D | %Rec | %Rec. Limits |
|----------------------------------------------------------|-------------|------------|---------------|------|---|------|--------------|
| 4:2 FTS                                                  | 37.4        | 33.0       |               | ng/L |   | 88   | 50 - 150     |
| 6:2 FTS                                                  | 37.9        | 29.0       |               | ng/L |   | 77   | 50 - 150     |
| 8:2 FTS                                                  | 38.3        | 27.8       |               | ng/L |   | 72   | 50 - 150     |
| N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)  | 40.0        | 36.5       |               | ng/L |   | 91   | 70 - 130     |
| N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA) | 40.0        | 35.9       |               | ng/L |   | 90   | 70 - 130     |
| Perfluorobutanesulfonic acid (PFBS)                      | 35.4        | 32.0       |               | ng/L |   | 90   | 70 - 130     |
| Perfluorobutanoic acid (PFBA)                            | 40.0        | 39.1       |               | ng/L |   | 98   | 50 - 150     |
| Perfluorodecanesulfonic acid (PFDS)                      | 38.6        | 36.6       |               | ng/L |   | 95   | 50 - 150     |
| Perfluorodecanoic acid (PFDA)                            | 40.0        | 39.3       |               | ng/L |   | 98   | 70 - 130     |
| Perfluorododecanoic acid (PFDoA)                         | 40.0        | 36.1       |               | ng/L |   | 90   | 70 - 130     |
| Perfluoroheptanesulfonic Acid (PFHpS)                    | 38.1        | 34.6       |               | ng/L |   | 91   | 50 - 150     |
| Perfluoroheptanoic acid (PFHpA)                          | 40.0        | 38.5       |               | ng/L |   | 96   | 70 - 130     |
| Perfluorohexanesulfonic acid (PFHxS)                     | 36.4        | 33.0       |               | ng/L |   | 91   | 70 - 130     |
| Perfluorohexanoic acid (PFHxA)                           | 40.0        | 37.3       |               | ng/L |   | 93   | 70 - 130     |
| Perfluorononanesulfonic acid (PFNS)                      | 38.4        | 34.7       |               | ng/L |   | 90   | 50 - 150     |
| Perfluorononanoic acid (PFNA)                            | 40.0        | 37.8       |               | ng/L |   | 95   | 70 - 130     |
| Perfluorooctanesulfonamide (FOSA)                        | 40.0        | 39.1       |               | ng/L |   | 98   | 50 - 150     |
| Perfluorooctane sulfonate (PFOS)                         | 37.1        | 34.7       |               | ng/L |   | 93   | 70 - 130     |
| Perfluorooctanoic acid (PFOA)                            | 40.0        | 38.9       |               | ng/L |   | 97   | 70 - 130     |
| Perfluoropentanesulfonic acid (PFPeS)                    | 37.5        | 35.7       |               | ng/L |   | 95   | 50 - 150     |
| Perfluoropentanoic acid (PFPeA)                          | 40.0        | 39.2       |               | ng/L |   | 98   | 50 - 150     |
| Perfluorotetradecanoic acid (PFTeA)                      | 40.0        | 45.8       |               | ng/L |   | 115  | 70 - 130     |
| Perfluorotridecanoic acid (PFTriA)                       | 40.0        | 36.6       |               | ng/L |   | 92   | 70 - 130     |
| Perfluoroundecanoic acid (PFUnA)                         | 40.0        | 38.6       |               | ng/L |   | 97   | 70 - 130     |

| Isotope Dilution | LCS %Recovery | LCS Qualifier | Limits   |
|------------------|---------------|---------------|----------|
| 13C8 FOSA        | 87            |               | 25 - 150 |
| 13C4 PFBA        | 98            |               | 25 - 150 |
| 13C2 PFDA        | 96            |               | 50 - 150 |
| 13C2 PFDoA       | 88            |               | 50 - 150 |
| 13C4 PFHpA       | 101           |               | 50 - 150 |
| 13C2 PFHxA       | 103           |               | 50 - 150 |
| 13C5 PFNA        | 97            |               | 50 - 150 |
| 13C4 PFOA        | 100           |               | 50 - 150 |
| 13C4 PFOS        | 110           |               | 50 - 150 |
| 13C5-PFPeA DNU   | 101           |               | 25 - 150 |
| 13C2 PFTeDA      | 75            |               | 50 - 150 |
| 13C2 PFUnA       | 85            |               | 50 - 150 |

Eurofins TestAmerica, Michigan

# QC Sample Results

Client: City of Cadillac Utilities  
Project/Site: WWTP Semiannual PFAS

Job ID: 190-21362-1

## Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCS 200-149141/2-A

Matrix: Water

Analysis Batch: 149190

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 149141

| Isotope Dilution | LCS       | LCS       | Limits   |
|------------------|-----------|-----------|----------|
|                  | %Recovery | Qualifier |          |
| d5-NEtFOSAA      | 90        |           | 50 - 150 |
| d3-NMeFOSAA      | 89        |           | 50 - 150 |
| M2-4:2 FTS       | 104       |           | 25 - 150 |
| M2-6:2 FTS       | 114       |           | 25 - 150 |
| M2-8:2 FTS       | 116       |           | 25 - 150 |
| 18O2 PFHxS       | 105       |           | 50 - 150 |

# Isotope Dilution Summary

Client: City of Cadillac Utilities  
Project/Site: WWTP Semiannual PFAS

Job ID: 190-21362-1

## Method: 537 (modified) - Fluorinated Alkyl Substances

Matrix: Water

Prep Type: Total/NA

### Percent Isotope Dilution Recovery (Acceptance Limits)

| Lab Sample ID      | Client Sample ID   | PFOSA<br>(25-150) | PFBA<br>(25-150) | PFDA<br>(50-150) | PFDoA<br>(50-150) | PFHpA<br>(50-150) | PFHxA<br>(50-150) | PFNA<br>(50-150) | PFOA<br>(50-150) |
|--------------------|--------------------|-------------------|------------------|------------------|-------------------|-------------------|-------------------|------------------|------------------|
| 190-21362-1        | Field Blank        | 92                | 102              | 102              | 92                | 105               | 105               | 104              | 101              |
| 190-21362-2        | Effluent           | 85                | 104              | 100              | 82                | 107               | 109               | 103              | 101              |
| 190-21362-3        | Effluent Duplicate | 78                | 100              | 88               | 77                | 102               | 104               | 90               | 97               |
| LCS 200-149141/2-A | Lab Control Sample | 87                | 98               | 96               | 88                | 101               | 103               | 97               | 100              |
| MB 200-149141/1-A  | Method Blank       | 71                | 100              | 102              | 77                | 102               | 104               | 106              | 107              |

### Percent Isotope Dilution Recovery (Acceptance Limits)

| Lab Sample ID      | Client Sample ID   | PFOS<br>(50-150) | PFPeA<br>(25-150) | PFTDA<br>(50-150) | PFUnA<br>(50-150) | d5-NEtFOSAA<br>(50-150) | d3-NMeFOSAA<br>(50-150) | M242FTS<br>(25-150) | M262FTS<br>(25-150) |
|--------------------|--------------------|------------------|-------------------|-------------------|-------------------|-------------------------|-------------------------|---------------------|---------------------|
| 190-21362-1        | Field Blank        | 111              | 105               | 84                | 93                | 86                      | 86                      | 113                 | 109                 |
| 190-21362-2        | Effluent           | 90               | 101               | 52                | 93                | 99                      | 83                      | 192 *               | 143                 |
| 190-21362-3        | Effluent Duplicate | 94               | 98                | 48 *              | 80                | 90                      | 89                      | 170 *               | 148                 |
| LCS 200-149141/2-A | Lab Control Sample | 110              | 101               | 75                | 85                | 90                      | 89                      | 104                 | 114                 |
| MB 200-149141/1-A  | Method Blank       | 109              | 106               | 67                | 87                | 84                      | 86                      | 104                 | 114                 |

### Percent Isotope Dilution Recovery (Acceptance Limits)

| Lab Sample ID      | Client Sample ID   | M282FTS<br>(25-150) | PFHxS<br>(50-150) |
|--------------------|--------------------|---------------------|-------------------|
| 190-21362-1        | Field Blank        | 118                 | 109               |
| 190-21362-2        | Effluent           | 162 *               | 89                |
| 190-21362-3        | Effluent Duplicate | 149                 | 94                |
| LCS 200-149141/2-A | Lab Control Sample | 116                 | 105               |
| MB 200-149141/1-A  | Method Blank       | 119                 | 104               |

### Surrogate Legend

PFOSA = 13C8 FOSA  
PFBA = 13C4 PFBA  
PFDA = 13C2 PFDA  
PFDoA = 13C2 PFDoA  
PFHpA = 13C4 PFHpA  
PFHxA = 13C2 PFHxA  
PFNA = 13C5 PFNA  
PFOA = 13C4 PFOA  
PFOS = 13C4 PFOS  
PFPeA = 13C5-PFPeA DNU  
PFTDA = 13C2 PFTeDA  
PFUnA = 13C2 PFUnA  
d5-NEtFOSAA = d5-NEtFOSAA  
d3-NMeFOSAA = d3-NMeFOSAA  
M242FTS = M2-4:2 FTS  
M262FTS = M2-6:2 FTS  
M282FTS = M2-8:2 FTS  
PFHxS = 18O2 PFHxS

## Definitions/Glossary

Client: City of Cadillac Utilities  
Project/Site: WWTP Semiannual PFAS

Job ID: 190-21362-1

### Qualifiers

#### LCMS

| Qualifier | Qualifier Description                                  |
|-----------|--------------------------------------------------------|
| *         | Isotope Dilution analyte is outside acceptance limits. |

### Glossary

| Abbreviation   | These commonly used abbreviations may or may not be present in this report.                                 |
|----------------|-------------------------------------------------------------------------------------------------------------|
| □              | Listed under the "D" column to designate that the result is reported on a dry weight basis                  |
| %R             | Percent Recovery                                                                                            |
| CFL            | Contains Free Liquid                                                                                        |
| CNF            | Contains No Free Liquid                                                                                     |
| DER            | Duplicate Error Ratio (normalized absolute difference)                                                      |
| Dil Fac        | Dilution Factor                                                                                             |
| DL             | Detection Limit (DoD/DOE)                                                                                   |
| DL, RA, RE, IN | Indicates a Dilution, Re-analysis, Re-extraction, or additional Initial metals/anion analysis of the sample |
| DLC            | Decision Level Concentration (Radiochemistry)                                                               |
| EDL            | Estimated Detection Limit (Dioxin)                                                                          |
| LOD            | Limit of Detection (DoD/DOE)                                                                                |
| LOQ            | Limit of Quantitation (DoD/DOE)                                                                             |
| MDA            | Minimum Detectable Activity (Radiochemistry)                                                                |
| MDC            | Minimum Detectable Concentration (Radiochemistry)                                                           |
| MDL            | Method Detection Limit                                                                                      |
| ML             | Minimum Level (Dioxin)                                                                                      |
| NC             | Not Calculated                                                                                              |
| ND             | Not Detected at the reporting limit (or MDL or EDL if shown)                                                |
| PQL            | Practical Quantitation Limit                                                                                |
| QC             | Quality Control                                                                                             |
| RER            | Relative Error Ratio (Radiochemistry)                                                                       |
| RL             | Reporting Limit or Requested Limit (Radiochemistry)                                                         |
| RPD            | Relative Percent Difference, a measure of the relative difference between two points                        |
| TEF            | Toxicity Equivalent Factor (Dioxin)                                                                         |
| TEQ            | Toxicity Equivalent Quotient (Dioxin)                                                                       |

## QC Association Summary

Client: City of Cadillac Utilities  
Project/Site: WWTP Semiannual PFAS

Job ID: 190-21362-1

### LCMS

#### Prep Batch: 149141

| Lab Sample ID      | Client Sample ID   | Prep Type | Matrix | Method | Prep Batch |
|--------------------|--------------------|-----------|--------|--------|------------|
| 190-21362-1        | Field Blank        | Total/NA  | Water  | 3535   |            |
| 190-21362-2        | Effluent           | Total/NA  | Water  | 3535   |            |
| 190-21362-3        | Effluent Duplicate | Total/NA  | Water  | 3535   |            |
| MB 200-149141/1-A  | Method Blank       | Total/NA  | Water  | 3535   |            |
| LCS 200-149141/2-A | Lab Control Sample | Total/NA  | Water  | 3535   |            |

#### Analysis Batch: 149190

| Lab Sample ID      | Client Sample ID   | Prep Type | Matrix | Method         | Prep Batch |
|--------------------|--------------------|-----------|--------|----------------|------------|
| 190-21362-2        | Effluent           | Total/NA  | Water  | 537 (modified) | 149141     |
| 190-21362-3        | Effluent Duplicate | Total/NA  | Water  | 537 (modified) | 149141     |
| MB 200-149141/1-A  | Method Blank       | Total/NA  | Water  | 537 (modified) | 149141     |
| LCS 200-149141/2-A | Lab Control Sample | Total/NA  | Water  | 537 (modified) | 149141     |

#### Analysis Batch: 149385

| Lab Sample ID | Client Sample ID | Prep Type | Matrix | Method         | Prep Batch |
|---------------|------------------|-----------|--------|----------------|------------|
| 190-21362-1   | Field Blank      | Total/NA  | Water  | 537 (modified) | 149141     |

# Lab Chronicle

Client: City of Cadillac Utilities  
Project/Site: WWTP Semiannual PFAS

Job ID: 190-21362-1

## Client Sample ID: Field Blank

Date Collected: 10/22/19 11:50

Date Received: 10/23/19 13:02

## Lab Sample ID: 190-21362-1

Matrix: Water

| Prep Type | Batch Type | Batch Method   | Run | Dilution Factor | Batch Number | Prepared or Analyzed | Analyst | Lab     |
|-----------|------------|----------------|-----|-----------------|--------------|----------------------|---------|---------|
| Total/NA  | Prep       | 3535           |     |                 | 149141       | 10/31/19 12:32       | MBM     | TAL BUR |
| Total/NA  | Analysis   | 537 (modified) |     | 1               | 149385       | 11/07/19 14:22       | JM1     | TAL BUR |

## Client Sample ID: Effluent

Date Collected: 10/22/19 11:55

Date Received: 10/23/19 13:02

## Lab Sample ID: 190-21362-2

Matrix: Water

| Prep Type | Batch Type | Batch Method   | Run | Dilution Factor | Batch Number | Prepared or Analyzed | Analyst | Lab     |
|-----------|------------|----------------|-----|-----------------|--------------|----------------------|---------|---------|
| Total/NA  | Prep       | 3535           |     |                 | 149141       | 10/31/19 12:32       | MBM     | TAL BUR |
| Total/NA  | Analysis   | 537 (modified) |     | 1               | 149190       | 11/02/19 00:36       | BWC     | TAL BUR |

## Client Sample ID: Effluent Duplicate

Date Collected: 10/22/19 11:55

Date Received: 10/23/19 13:02

## Lab Sample ID: 190-21362-3

Matrix: Water

| Prep Type | Batch Type | Batch Method   | Run | Dilution Factor | Batch Number | Prepared or Analyzed | Analyst | Lab     |
|-----------|------------|----------------|-----|-----------------|--------------|----------------------|---------|---------|
| Total/NA  | Prep       | 3535           |     |                 | 149141       | 10/31/19 12:32       | MBM     | TAL BUR |
| Total/NA  | Analysis   | 537 (modified) |     | 1               | 149190       | 11/02/19 00:44       | BWC     | TAL BUR |

### Laboratory References:

TAL BUR = Eurofins TestAmerica, Burlington, 30 Community Drive, Suite 11, South Burlington, VT 05403, TEL (802)660-1990

### Analyst References:

Lab: TAL BUR

Batch Type: Prep

MBM = Morgan Morrill

Batch Type: Analysis

BWC = Bradley Chirgwin

JM1 = John Murray

## Accreditation/Certification Summary

Client: City of Cadillac Utilities  
Project/Site: WWTP Semiannual PFAS

Job ID: 190-21362-1

### Laboratory: Eurofins TestAmerica, Michigan

All accreditations/certifications held by this laboratory are listed. Not all accreditations/certifications are applicable to this report.

| Authority | Program | Identification Number | Expiration Date |
|-----------|---------|-----------------------|-----------------|
| Michigan  | State   | 0057                  | 05-05-20        |

### Laboratory: Eurofins TestAmerica, Burlington

All accreditations/certifications held by this laboratory are listed. Not all accreditations/certifications are applicable to this report.

| Authority                         | Program               | Identification Number | Expiration Date |
|-----------------------------------|-----------------------|-----------------------|-----------------|
| ANAB                              | Dept. of Defense ELAP | L2336                 | 02-25-20        |
| Connecticut                       | State                 | PH-0751               | 09-30-21        |
| Connecticut                       | State Program         | PH-0751               | 09-30-21        |
| DE Haz. Subst. Cleanup Act (HSCA) | State                 | N/A                   | 05-15-20        |
| Florida                           | NELAP                 | E87467                | 06-30-20        |
| Minnesota                         | NELAP                 | 050-999-436           | 12-31-19        |
| New Hampshire                     | NELAP                 | 2006                  | 12-18-19        |
| New Jersey                        | NELAP                 | VT972                 | 06-30-20        |
| New York                          | NELAP                 | 10391                 | 03-31-20        |
| Pennsylvania                      | NELAP                 | 68-00489              | 04-30-20        |
| Rhode Island                      | State                 | LAO00298              | 12-30-19        |
| Rhode Island                      | State Program         | LAO00298              | 12-30-19        |
| US Fish & Wildlife                | US Federal Programs   | 058448                | 07-31-20        |
| USDA                              | US Federal Programs   | P330-17-00272         | 08-09-20        |
| Vermont                           | State                 | VT4000                | 12-31-19        |
| Virginia                          | NELAP                 | 460209                | 12-14-19        |

Phone: 810-229-2763 Fax: 810-229-0000

Ver: 01/16/2019



Environment Testing  
TestAmerica

☐ SDS or Known Hazard Information Supplied by Client

☐ Discrepancies

☐ Short Hold

☐ Rush ☐ 24hr ☐ 2day ☐ 3day ☐ 5day ☐ Other

Client ID City of Cadillac

Work Order # 198-21362

## Cooler/Sample Receipt

(AFTER HOURS receipt, complete gray areas)  
Place cooler in walk-in place in this form in Receiving in-  
box. Date/Time received Initials

Receipt evaluation performed by - Initials AmT Date 10/23/19 Time 13:42

### Method of Shipment:

☐ Walk-In Client ☐ TestAmerica Field/Courier

☐ Other Client/3<sup>rd</sup> Party Courier

☐ Fed Ex Tracking #

☒ UPS Tracking # 1Z E47 491 03 5491 5845

☐ Other Ground

### Shipping Container Type:

☐ Cooler ☐ Box Insulated

☐ None ☒ Other Shipped

### Packing Materials:

☒ Plastic Bags ☐ Foam

☒ Bubble Wrap ☐ Paper

☐ Packing Peanuts ☐ None

☐ Other

### Custody Seals Intact:

☒ Yes 4/8/19/19 ☐ No

☐ N/A (not used or required)

### Cooling Materials:

☒ Ice (solid) ☐ Ice (Melted)

☒ Blue Ice ☐ None

☐ Other

Biological: Temp (°C) Corrected  
Samples

Frozen  
yes no

Received within 2 hours  
yes no

Sample Flagged  
yes no

### Receipt Temperatures

| Thermometer ID | Observed (°C) | Corrected (°C) | Blank                    | Temp                                | Received on same day sampled?                                       | Acceptable?                                                         | Cooler ID | Note Affected Samples if temperature not acceptable |
|----------------|---------------|----------------|--------------------------|-------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------|-----------|-----------------------------------------------------|
| 140252483      |               |                | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> Yes <input type="checkbox"/> No            | <input type="checkbox"/> Yes <input type="checkbox"/> No            |           |                                                     |
| 140252476      |               |                | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> Yes <input type="checkbox"/> No            | <input type="checkbox"/> Yes <input type="checkbox"/> No            |           |                                                     |
| CP313207       | 2.1           | 2.2            | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> Yes <input type="checkbox"/> No | <input checked="" type="checkbox"/> Yes <input type="checkbox"/> No |           |                                                     |

\* Receipt temperatures are considered acceptable if the samples are received on the same day they were collected & show signs that the cooling process has started. Temperature acceptance for most tests is ≤6.0°C, but not frozen.

| Receipt Questions**                                                                                                                                                                                              | Y                                   | N | n/a                                 | "No" answers require additional comment          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|---|-------------------------------------|--------------------------------------------------|
| COC present & TA receipt signature, date, & time properly documented?                                                                                                                                            | <input checked="" type="checkbox"/> |   |                                     |                                                  |
| Containers & labels in good condition? (unbroken, not leaking, appropriately filled, labels legible & attached)                                                                                                  | <input checked="" type="checkbox"/> |   |                                     |                                                  |
| Appropriate containers used & adequate volume provided?                                                                                                                                                          | <input checked="" type="checkbox"/> |   |                                     | Preserved Bottles Checked with pH Strips* Yes No |
| Number of sample containers match COC?                                                                                                                                                                           | <input checked="" type="checkbox"/> |   |                                     |                                                  |
| Samples received within hold time?                                                                                                                                                                               | <input checked="" type="checkbox"/> |   |                                     |                                                  |
| Samples submitted for GRO and Volatiles analyses (8260, 624, 524) received without headspace?                                                                                                                    |                                     |   | <input checked="" type="checkbox"/> |                                                  |
| Was a Trip Blank received with VOA samples?                                                                                                                                                                      |                                     |   | <input checked="" type="checkbox"/> |                                                  |
| Were the samples free of any questionable physical conformities? For example, field duplicates or multiple bottles of the same sample do not significantly vary in appearance (color, proportion of solids, etc) | <input checked="" type="checkbox"/> |   |                                     |                                                  |
| Were the COC bottle labels and all other items free of all other discrepancies or issues that would need to be addressed with the Project Manager and/or Client?                                                 | <input checked="" type="checkbox"/> |   |                                     |                                                  |

\*\* May not be applicable if samples are not for compliance testing

\* Excludes FOG, Volatiles, TOC, Vials

### Client Contact Record

Contact via: ☐ Phone ☐ Email ☐ Other Person Contacted: Date/Time:

☐ Discrepancy allowance agreement is on record in the client project file

Discussion/Resolution:

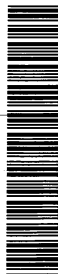
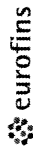
Any additional documentation and clarification from client must be noted in the narrative and/or scanned into the COC directory

Reviewed by PMS Signature Date

WI Page 1 of 1

WI No: DT-SCA-WI-001-12  
effective 06/11/12

## Chain of Custody Record



|                                                                                           |  |                                            |  |                                                        |  |                                     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |
|-------------------------------------------------------------------------------------------|--|--------------------------------------------|--|--------------------------------------------------------|--|-------------------------------------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Client Information (Sub Contract Lab)</b>                                              |  | Sampler: Schaefer, Sue                     |  | Lab PM: Schaefer, Sue                                  |  | Carrier Tracking No(s): 190-24973.1 |  | COC No: 190-24973.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |
| Client Contact: Shipping/Receiving                                                        |  | Phone:                                     |  | E-Mail: sue.schafer@testamericainc.com                 |  | State of Origin: Michigan           |  | Page: 1 of 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |
| Company: TestAmerica Laboratories, Inc.                                                   |  | Due Date Requested: 11/8/2019              |  | TAT Requested (days):                                  |  | Job #: 190-21362-1                  |  | Preservation Codes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |
| Address: 30 Community Drive, Suite 11, South Burlington                                   |  | PO #:                                      |  | WO #:                                                  |  | Project #:                          |  | SSOW#:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
| State, Zip: VT, 05403                                                                     |  | Phone: 802-660-1990(Tel) 802-660-1919(Fax) |  | Email:                                                 |  | Project Name: WWTP Semiannual PFAS  |  | Site:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
| Matrix (W=water, S=solid, O=waste/oil, BT=Tissue, A=Air)                                  |  | Sample Type (C=Comp, G=grab)               |  | Sample Time                                            |  | Sample Date                         |  | Sample Identification - Client ID (Lab ID)                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| Field Blank (190-21362-1)                                                                 |  | Water                                      |  | 11:50 Eastern                                          |  | 10/22/19                            |  | Field Filtered Sample (Yes or No)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
| Effluent (190-21362-2)                                                                    |  | Water                                      |  | 11:55 Eastern                                          |  | 10/22/19                            |  | Perform MS/MSD (Yes or No)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| Effluent Duplicate (190-21362-3)                                                          |  | Water                                      |  | 11:55 Eastern                                          |  | 10/22/19                            |  | PFAS ID/3535 JWVT (MOD) PFAS, Standard List (24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
|                                                                                           |  |                                            |  |                                                        |  |                                     |  | Total Number of Containers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
|                                                                                           |  |                                            |  |                                                        |  |                                     |  | Special Instructions/Note:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
|                                                                                           |  |                                            |  |                                                        |  |                                     |  | warded under chain-of-custody. If the laboratory does not to accreditation status should be brought to TestAmerica                                                                                                                                                                                                                                                                                                                                                                                                   |  |
|                                                                                           |  |                                            |  |                                                        |  |                                     |  | 190-21362 Chain of Custody                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
|                                                                                           |  |                                            |  |                                                        |  |                                     |  | Note: Since laboratory accreditations are subject to change, TestAmerica Laboratories, Inc. places the ownership of method, analyte & accreditation cor currently maintain accreditation in the State of Origin listed above for analysis/tests/matrix being analyzed, the samples must be shipped back to the Test Laboratories, Inc. attention immediately. If all requested accreditations are current to date, return the signed Chain of Custody attesting to said compliance to TestAmerica Laboratories, Inc. |  |
| <b>Possible Hazard Identification</b>                                                     |  | Unconfirmed                                |  | Deliverable Requested: I, II, III, IV, Other (specify) |  | Primary Deliverable Rank: 2         |  | Sample Disposal (A fee may be assessed if samples are retained longer than 1 month)                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |
|                                                                                           |  |                                            |  |                                                        |  |                                     |  | Return To Client <input type="checkbox"/> Disposal By Lab <input type="checkbox"/> Archive For <input type="checkbox"/> Months                                                                                                                                                                                                                                                                                                                                                                                       |  |
| Empty Kit Relinquished by:                                                                |  | Date:                                      |  | Time:                                                  |  | Method of Shipment:                 |  | Special Instructions/QC Requirements:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
| Relinquished by: <i>ASW</i>                                                               |  | Date/Time: 10-23-19 1500                   |  | Company: ETA                                           |  | Received by: <i>ASW</i>             |  | Date/Time: 10/24/19 10:40                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |
| Relinquished by:                                                                          |  | Date/Time:                                 |  | Company:                                               |  | Received by:                        |  | Date/Time:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| Relinquished by:                                                                          |  | Date/Time:                                 |  | Company:                                               |  | Received by:                        |  | Date/Time:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| Custody Seals Intact: <input checked="" type="checkbox"/> Yes <input type="checkbox"/> No |  | Custody Seal No.: 1055019                  |  | Cooler Temperature(s) °C and Other Remarks: 0-7        |  |                                     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |

Do Not Lift Using This Tag

**TestAmerica**

THE LEADER IN ENVIRONMENTAL TESTING

ORIGIN ID:DEQA (810) 229-2763  
SHIPPING DEPARTMENT  
TESTAMERICA  
10448 CITATION DRIVE  
SUITE 200  
BRIGHTON, MI 48116  
UNITED STATES US

SHIP DATE: 23OCT19  
ACTWGT: 39.75 LB  
CAD: 0183192/CAFE3211

BILL RECIPIENT

TO  
SAMPLE RECIEVING  
TEST AMERICA BURLINGTON  
30 COMMUNITY DR  
STE 11  
SOUTH BURLINGTON VT 054036834

(802) 660-1990  
INV:  
PO:

REF:

DEPT:



FedEx  
Express



TRK#  
0201 4506 9512 0121

THU - 24 OCT 10:30A  
PRIORITY OVERNIGHT

**NL BTVA**

05403

VT-US BTV



Part # 159469-434 RPT EXP 07/20

3601/2622/23155